Course Project- CIE 237- Spring 2026

Introduction

The problem of recovering signals corrupted by noise is a key problem in digital communications. The correlation receiver can be shown to be the best solution (optimal) for recovering a known waveform in the presence of **a**dditive **w**hite **G**aussian **n**oise (AWGN). A correlation receiver compares the received signal to a set of waveforms it expects to receive. This scenario applies directly to a digital communication problem since each bit (or bit sequence) can be encoded to a set of distinct waveforms known to both the transmitter and receiver. The probability of detection increases with increasing the Signal-to-Noise-Ratio (SNR).

A **correlation receiver** parses the signal into intervals that are synchronized with the bit intervals and integrates the product of the received signal and the templates over the interval to produce correlation values. The correlation values become detection statistics used to decide the most likely bit value that was transmitted for each interval.

A bit error occurs when noise or bandwidth limitations result in an incorrect decision. For a given test or experiment, **the numbers of errors per bit is referred to as the bit error rate (BER).**

The correlation receiver slides the template over the data, estimating the degree of the match between the incoming signal and the template. **This operation is similar to convolution with a subtle difference;** convolution reverses the system impulse response (in time) prior to performing the multiplication and integration over the sliding window. Therefore, if a template is flipped (or time-reversed) and treated as a filter's impulse response, then the filter operation reverses it back to the original template and effectively performs a correlation. The resulting operation, referred to as **matched filtering**, has an identical outcome to the correlation receiver.

- You are required to understand the design principles of a matched filter and the mathematics therein and use MATLAB Simulink to simulate a digital channel producing a graph that depicts the relation of the bit error rate and the signal to noise ratio. You should consider 2-level pulse amplitude modulation (PAM) and more generally M-levels PAM and provide the symbol error probability in both cases.

- Repeat previous simulations using MATLAB codes instead of Simulink and compare the SNRS results for different AWGN values and for different PAM pulses. Comment on your results.

- You are required to explain all figures and results in your report.

- You are required to provide the citation of all referenced work, figures, books....... etc

Project Regulations

- If you want to work in pairs (each team contains 2 students), that is acceptable. However, you must inform the instructor of your intention by March 10, 2026. After that time, you will be required to submit an individual report. Collaborative reports must indicate the authorship of each section.

- Your report should be less than 15 pages including figures, comments, and references. It should be typed with machine generated graphics. The title page need not be included in the page count. Your sources must be properly referenced.

- Deadline for the final report submission is May 15, 2026.

- Each team will present his project during the last lecture time. The discussion will be held in the last week of the semester.

- You are not allowed to discuss your work with anyone other than the instructor.

Project Grading

Grading will done on the basis of:

- Technical content of the report.

- Project final presentation, including presentation graphics and appeal, accuracy and comprehensibility of presentation content, audience capturing, and audience participation and reply to questions

- Insight: how detailed and perceptive is your analysis?

- How effectively you use the literature? Can I identity the sources of your information well enough to reproduce your results?

- Clarity and writing. The report should be written in such a way that a technically competent person could reproduce your results if he/she wanted to

- Define all acronyms and describe any assumptions that you make. You should be able to pick up this paper a year from now and be able to read it.

- Your course project will be 15 % of your total grade. Marks will be assigned on

    1. Planning (10%)
    2. Final report (40%)
    3. Final presentation (50%)